

Local AI System – Project Description

Prototype & Practice Project (Local, Human-in-the-Loop AI)

1. Project Theme

This project focuses on building a local, enterprise-style AI system that assists humans in working with large volumes of internal documents while preserving full data sovereignty. The AI acts strictly as a supporting assistant and never as a decision-maker.

2. Core Principles

- Human authority is final; AI suggestions must always be reviewed.
- AI outputs are probabilistic and may contain errors or hallucinations.
- The system runs fully on local hardware without cloud dependency.
- Traceability and transparency are mandatory.

3. Data Provided by Users

Users provide internal enterprise data such as policies, specifications, contracts, and training materials. These documents form the knowledge base of the system.

- Internal guidelines, standards, and policies
- Functional specifications (multiple versions/releases)
- Contracts and legal documents
- Training and onboarding materials

4. What Users Expect

- Fast access to relevant information across many documents
- Answers supported by document references (source, section, page)
- Drafts and summaries that reduce manual work
- Clear indication when information cannot be found

5. Source Attribution Requirement

For all document-based questions, the system must provide clear source references. This includes document name, section or heading, and page or paragraph where possible. This enables verification and reduces risk.

6. What the System Is Not

- Not an autonomous decision-making system

- Not a cloud-based AI service
- Not a replacement for employees or experts
- Not a system that guarantees correctness or business success

7. Prototype Goal

The goal of this prototype is learning and architectural understanding. It demonstrates the full AI pipeline at a small scale, from data ingestion to retrieval and answer generation, while staying aligned with enterprise constraints.